Daily Coal Stock Monitoring and Consumption Analysis

1.Project Overview

This project focuses on analyzing daily coal inventory data to ensure efficient stock management and uninterrupted plant operations. Coal is a critical resource for power generation and industrial production, and maintaining the right stock level is essential to avoid operational disruptions or excessive holding costs.

The dataset includes key operational parameters such as daily coal receipt, daily consumption, actual stock levels, and normative (required) stock levels. By comparing actual stock with predefined normative benchmarks, the project evaluates whether inventory levels are within safe operating limits.

1.1 Project Objective:

The main objective of this project is to analyze daily coal inventory data to ensure optimal stock levels, prevent shortages, and improve operational efficiency through effective monitoring and data-driven decision-making.

- **Monitor Coal Supply and Usage**
  - Track daily coal receipt and daily consumption.
  - Identify fluctuations in supply and demand.
- **Evaluate Stock Against Normative Levels**
  - Compare actual stock with predefined normative (required) stock.
  - Measure deviation using Actual vs Normative %.
- **Identify Risk Conditions**
  - Detect potential coal shortages that may impact operations.
  - Identify excess stock that may increase storage and holding costs.
- **Analyze Trends and Patterns**
  - Study daily and monthly consumption trends.
  - Understand stock movement over time.
- **Improve Inventory Planning**
  - Support forecasting of coal requirements.
  - Assist management in better procurement decisions.
- **Develop a Monitoring Dashboard**
  - Create interactive visual reports (Power BI / Excel).

2.Source Description and Timeline:

The dataset is obtained from India Data Portal. This source is from Ministry of power.Coal stocks data is taken from thermal power stations across India.The data is collected daily.It contains coal stock data by state,sector,Individual power station.

The data field includes mode of transport,capacity,normative stocks,actual stocks,receipt,consumption,plant load factor,the criticalness of the plant,reason for critical ,etc.,for each station.

Source link: https://indiadataportal.com/p/power/r/mop-coal_stock-pl-dl-aaa
Time Line: 2018-2025

Domain: Energy

3.Problem Statement:

If coal stock falls below the **normative (required) level**, it may result in operational risks such as reduced output or plant shutdown. On the other hand, excess stock beyond the required level can increase storage costs and reduce inventory efficiency.

Currently, there is a need to systematically monitor and analyze daily coal inventory data to:

- Track variations in receipt and consumption
- Compare actual stock with normative stock levels
- Identify stock shortages or excess conditions
- Measure stock performance using key indicators
- Support effective procurement and inventory planning decisions

Therefore, this project aims to analyze daily coal stock data to identify deviations from normative levels, assess inventory performance, and provide actionable insights for improved stock management and operational efficiency.

4.Attribute (Column /Features) Details:

Attribute Name	Description	Data type
Id	Unique identifier assigned to each record/row in the dataset	Integer
Date	The specific day on which coal data is recorded	Date
Statename	Name of the state where the plant or coal stock location is situated	Text
Statecode	Numeric code representing each state	Text
Powerstation Name	Name of the thermal power plant where coal stock data is recorded	Text
Sector	Category under which the power station operates (e.g., Central, State, Private)	Text
Utility	Name of the electricity generating company or organization operating the power station	Text
Mode of Transport	The method used to transport coal from the source to the power station (e.g., Rail, Road, Sea)	Text

Capacity	The maximum electricity generation capacity of the power station, usually measured in Megawatts (MW)	Decimal
Daily requirement	The estimated quantity of coal required per day to operate the power station efficiently	Decimal
Daily Consumption	Quantity of coal actually consumed by the power station on a particular day	Decimal
Required Normative stock	The minimum stock level required as per guidelines to ensure uninterrupted operations	Decimal
Normative stock days	Number of days the normative stock should cover based on daily requirement	Decimal
Indigenous Stock	Quantity of coal available from domestic (within country) sources	Decimal
Import Stock	Quantity of coal available from imported sources	Decimal
Total Stock	Total available coal stock (Indigenous + Import Stock)	Integer
Stock days	Number of days the current total stock can sustain operations based on daily consumption	Integer
Plf_prcnt	Percentage of actual power generation compared to maximum possible generation capacity	Decimal (Percentage)
Actual_vs normative_stock _prcnt	Percentage comparison of Actual Stock with Required Normative Stock	Decimal (Percentage)
Is critical	Indicates whether the stock level is below the critical threshold	Boolean (Yes/No) or Text
Remarks	Additional comments or observations regarding stock condition	Text

5. Tools & Technologies :

● Excel: Data cleaning, transformation, and Pivot Tables.

● Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

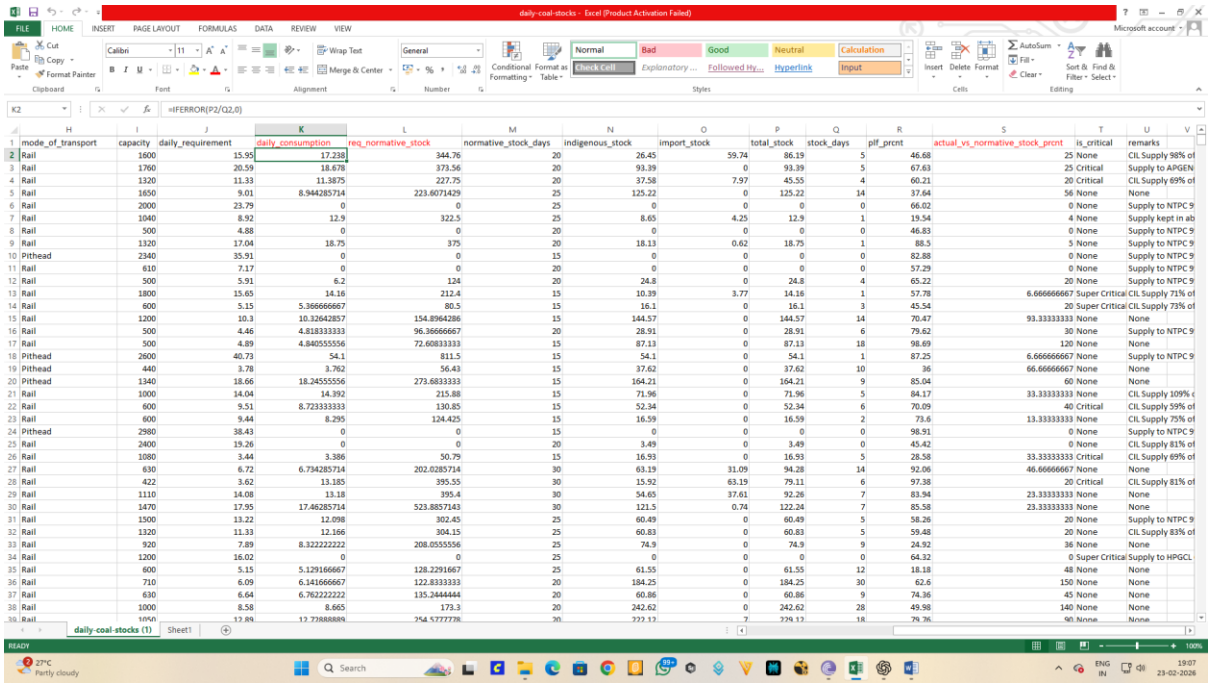
Tasks Performed:

● Data Cleaning & Transformation:

Data was cleaned and Removed duplicates, handled missing values, standardized formats, and created calculated fields.

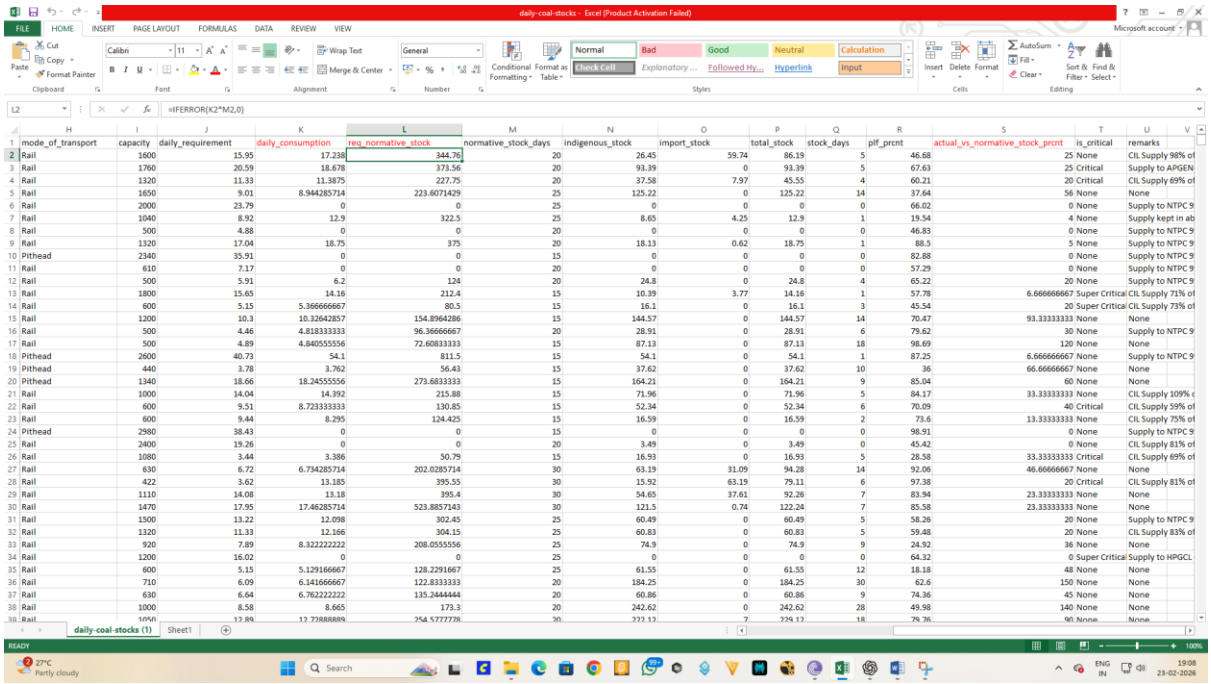
Daily consumption , Req_normative_stock and Actual vs Normative percentage values are calculated.

Daily_consumption=Total Stock/Stock days



mode_of_transport	capacity	daily_requirement	daily_consumption	req_normative_stock	normative_stock_days	indigenous_stock	import_stock	total_stock	stock_days	plf_prcnt	actual_vs_normative_stock_prcnt	is_critical	remarks
Rail	1600	15.95	17.238	344.76	20	26.45	59.74	86.19	5	46.68	25	None	CIL Supply 98% of
Rail	1760	20.59	18.678	373.56	20	93.39	0	93.39	5	67.63	25	Critical	Supply to APGEN
Rail	1320	11.33	11.3875	227.75	20	37.58	7.97	45.55	4	60.21	20	Critical	CIL Supply 69% of
Rail	1650	9.01	8.944285714	223.6071429	25	125.22	0	125.22	14	37.64	36	None	None
Rail	2000	23.79	0	0	25	0	0	0	0	66.02	0	None	Supply to NTPC 9
Rail	1040	8.92	12.9	322.5	25	8.65	4.25	12.9	1	19.54	4	None	Supply kept in ab
Rail	500	4.88	0	0	20	0	0	0	0	46.83	0	None	Supply to NTPC 9
Rail	1320	17.04	18.75	375	20	18.13	0.62	18.75	1	88.5	5	None	Supply to NTPC 9
Pithead	2340	35.91	0	0	15	0	0	0	0	82.88	0	None	Supply to NTPC 9
Rail	610	7.17	0	0	20	0	0	0	0	57.29	0	None	Supply to NTPC 9
Rail	500	5.91	6.2	124	20	24.8	0	24.8	4	65.22	20	None	Supply to NTPC 9
Rail	1800	15.65	14.16	212.4	15	10.39	3.77	14.16	1	57.78	6.666666667	Super Critical	CIL Supply 71% of
Rail	600	5.15	5.366666667	80.5	15	16.1	0	16.1	3	45.54	20	Super Critical	CIL Supply 73% of
Rail	1200	10.3	10.32642857	154.8964286	15	144.57	0	144.57	14	70.47	93.33333333	None	None
Rail	500	4.46	4.818333333	96.36666667	20	28.91	0	28.91	6	79.62	30	None	Supply to NTPC 9
Rail	500	4.89	4.840555556	72.60833333	15	87.13	0	87.13	18	98.69	120	None	None
Pithead	2600	40.73	54.1	811.5	15	54.1	0	54.1	1	87.25	6.666666667	None	Supply to NTPC 9
Pithead	440	3.78	3.762	56.43	15	37.62	0	37.62	10	36	66.66666667	None	None
Pithead	1340	18.66	18.24555556	273.6833333	15	164.21	0	164.21	9	85.04	60	None	None
Rail	1000	14.04	14.392	215.88	15	71.96	0	71.96	5	84.17	33.33333333	None	CIL Supply 109%
Rail	600	9.51	8.723333333	130.85	15	52.34	0	52.34	6	70.09	40	Critical	CIL Supply 59% of
Rail	600	9.44	8.295	124.425	15	16.59	0	16.59	2	73.6	13.33333333	None	CIL Supply 75% of
Pithead	2980	38.43	0	0	15	0	0	0	0	98.91	0	None	Supply to NTPC 9
Rail	2400	19.26	0	0	20	3.49	0	3.49	0	45.42	0	None	CIL Supply 81% of
Rail	1080	3.44	3.386	50.79	15	16.93	0	16.93	5	28.58	33.33333333	Critical	CIL Supply 69% of
Rail	630	6.72	6.734285714	202.0285714	30	63.19	31.09	94.28	14	92.06	46.66666667	None	None
Rail	422	3.62	13.185	395.55	30	15.92	63.19	79.11	6	97.38	20	Critical	CIL Supply 81% of
Rail	1110	14.08	13.18	395.4	30	54.65	37.61	92.26	7	83.94	23.33333333	None	None
Rail	1470	17.95	17.46285714	523.8857143	30	121.5	0.74	122.24	7	85.58	23.33333333	None	None
Rail	1500	13.22	12.098	302.45	25	60.49	0	60.49	5	58.26	20	None	Supply to NTPC 9
Rail	1320	11.33	12.166	304.15	25	60.83	0	60.83	5	59.48	20	None	CIL Supply 83% of
Rail	920	7.89	8.322222222	208.0555556	25	74.9	0	74.9	9	24.92	36	None	None
Rail	1200	16.02	0	0	25	0	0	0	0	64.32	0	Super Critical	Supply to HPGCL
Rail	600	5.15	5.129166667	128.2291667	25	61.55	0	61.55	12	18.18	48	None	None
Rail	710	6.09	6.141666667	122.8333333	20	184.25	0	184.25	30	62.6	150	None	None
Rail	630	6.64	6.762222222	135.2444444	20	60.86	0	60.86	9	74.36	45	None	None
Rail	1000	8.58	8.665	173.3	20	242.62	0	242.62	28	49.98	140	None	None
Rail	1050	12.89	12.7888889	254.5777778	20	222.12	7	229.12	18	79.26	90	None	None

Req_normative_stock =Daily Requirement * Normative Days



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Rail	1320	11.33	11.3875	227.75	20	37.58	7.97	45.55	4	60.21	20	Critical	CIL Supply 69% of
Rail	1650	9.01	8.944285714	223.6071429	25	125.22	0	125.22	14	37.64	56	None	None
Rail	2000	23.79	0	0	25	0	0	0	0	66.02	0	None	Supply to NTPC 9
Rail	1040	8.92	12.9	322.5	25	8.65	4.25	12.9	1	19.54	4	None	Supply kept in ab
Rail	500	4.88	0	0	20	0	0	0	0	46.83	0	None	Supply to NTPC 9
Rail	1320	17.04	18.75	375	20	18.13	0.62	18.75	1	88.5	5	None	Supply to NTPC 9
Pithead	2340	35.91	0	0	15	0	0	0	0	82.88	0	None	Supply to NTPC 9
Rail	610	7.17	0	0	20	0	0	0	0	57.29	0	None	Supply to NTPC 9
Rail	500	5.91	6.2	124	20	24.8	0	24.8	4	65.22	20	None	Supply to NTPC 9
Rail	1800	15.65	14.16	212.4	15	10.39	3.77	14.16	1	57.78	6.666666667	Super Critical	CIL Supply 71% of
Rail	600	5.15	5.366666667	80.5	15	16.1	0	16.1	3	45.54	20	Super Critical	CIL Supply 73% of
Rail	1200	10.3	10.32642857	154.8964286	15	144.57	0	144.57	14	70.47	93.33333333	None	None
Rail	500	4.46	4.818333333	96.36666667	20	28.91	0	28.91	6	79.62	30	None	Supply to NTPC 9
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Pithead	2600	40.73	54.1	811.5	15	54.1	0	54.1	1	87.25	6.666666667	None	Supply to NTPC 9
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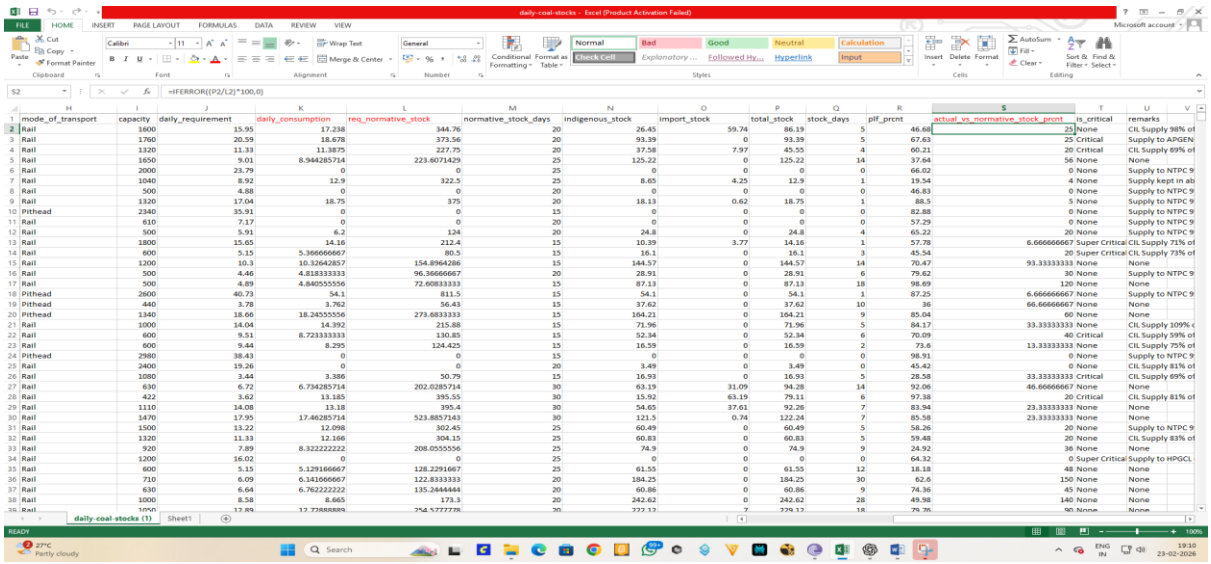
Rail	630	6.64	6.762222222	135.2444444	20	60.86	0	60.86	9	74.36	45	None	None
Rail	1000	8.58	8.665	173.3	20	242.62	0	242.62	28	49.98	140	None	None
Rail	1050	12.89	12.73888889	254.5777778	20	222.12	7	229.12	18	79.76	90	None	None

Actual vs Normative percentage=(Total Stock/Req. Normative Stock)*100.



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Pithead	2980	38.43	0	0	15	0	0	0	0	98.91	0	None	Supply to NTPC 9
Rail	2400	19.26	0	0	20	3.49	0	3.49	0	45.42	0	None	CIL Supply 81% of
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Rail	1110	14.08	13.18	395.4	30	54.65	37.61	92.26	7	83.94	23.33333333	None	None
Rail	1470	17.95	17.46285714	523.8857143	30	121.5	0.74	122.24	7	85.58	23.33333333	None	None
Rail	1500	13.22	12.098	302.45	25	60.49	0	60.49	5	58.26	20	None	Supply to NTPC 9
Rail	1320	11.33	12.166	304.15	25	60.83	0	60.83	5	59.48	20	None	CIL Supply 83% of
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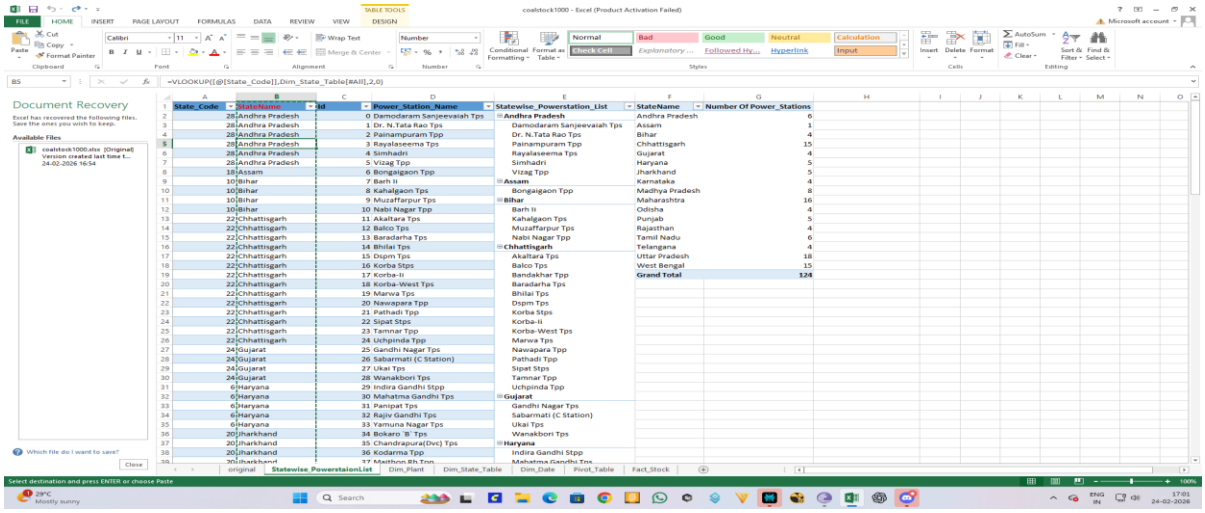
Rail	630	6.64	6.762222222	135.2444444	20	60.86	0	60.86	9	74.36	45	None	None
Rail	1000	8.58	8.665	173.3	20	242.62	0	242.62	28	49.98	140	None	None
Rail	1050	12.89	12.73888889	254.5777778	20	222.12	7	229.12	18	79.76	90	None	None

● Filtering & Sorting: Organized data to focus on relevant records.

→Duplicate State names are removed and Separate **Dim_State table** is created with unique State code and State name.

→Duplicate PowerStation names were removed and separate **Statewise PowerStation_List** table is creatd.Statenames of Powerstation is retrieved with vlookup function. Statewise PowerStation_List consists of State_Code,State_Name, Id,Powerstation_Name.The Original dataset contains duplicate powerstation names as it shows the daily transaction.

=VLOOKUP([@[State_Code]],Dim_State_Table[#All],2,0)



• Convert the data into Fact and Dimension Table

Dim_Plant

Plant ID	Plant Name	State	Sector	Utility	Capacity

Dim_Date

Date ID	Date	Month	Year	Quarter

Fact_Stock

Plant ID	Date ID	Mode of Transport	Daily Consumption	Daily Requirement

Indigenous Stock	Import Stock	Stock Days	Normative Stock Days	Total Stock

Req. Normative Stock	PLF%	Actual vs Normative Stock	Is critical	Remarks

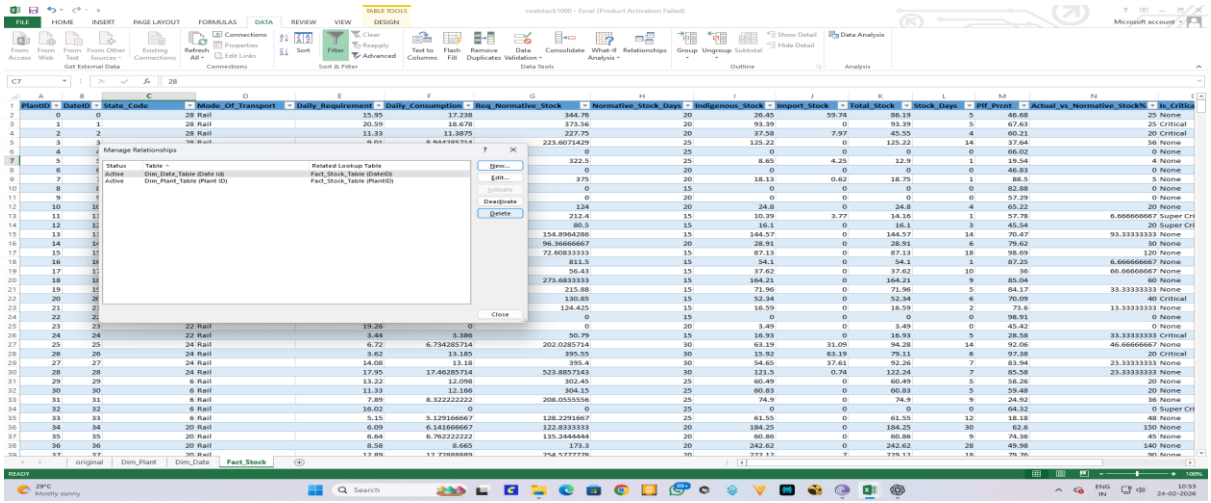
Relationship is created between 3 tables.

Table **Related Lookup Table**

Dim_Date_Table(DateID) →Fact_Stock_Table(Date_ID)

Dim_Plant_Table(Plant_ID) →Fact_Stock_Table(Plant_ID)

Fact_Stock_Table(State_Code) →Dim_State_Table(State_Code



● Pivot Tables: Generated Pivot Tables for data summarisation and initial insights.

The following pivot table depicts statewise total of Indigenous ,Import and Total stocks.

State_Name	Total Indigenous_Stock	Total Import_Stock	Total_Stock
Andhra Pradesh	3394.56	669.18	4063.74
Assam	22	0	22
Bihar	627.61	5.58	633.17
Chhattisgarh	5964.32	14.12	5978.44
Gujarat	2213.06	1061.89	3274.95
Haryana	1956.42	0	1956.42
Jharkhand	6415.35	73.86	6489.21
Karnataka	2007.28	0	2007.28
Madhya Pradesh	15236.89	239.73	15476.59
Maharashtra	7545.86	104.05	7649.91
Odisha	3817.75	135.7	3953.47
Punjab	3222.84	182.57	3405.4
Rajasthan	4652.33	82.29	4734.62
Tamil Nadu	2940.42	1906.81	4847.22
Telangana	10662.12	0	10662.12
Uttar Pradesh	25627.95	0	25627.95
West Bengal	4484.17	2	4486.17
Grand Total	**100790.93**	**4477.78**	**105268.66**

The following pivot table depicts Total Requiement and Total consumption

State_Name	Total Daily_Requirement	Total Daily_Consumption
Andhra Pradesh	818.27	596.95
Assam	41.56	21.56
Bihar	583.27	221.23
Chhattisgarh	1605.41	1179.32
Gujarat	337.43	403.38
Haryana	427.14	303.73
Jharkhand	314.3	313.77
Karnataka	421.44	405.36
Madhya Pradesh	1236.86	1162.03
Maharashtra	1748.5	1726.07
Odisha	628.35	637.21
Punjab	435.53	434.41
Rajasthan	411.73	401.61
Tamil Nadu	561.42	578.32
Telangana	564.55	564.99
Uttar Pradesh	1783.8	1793.44
West Bengal	1154.68	951.94
Grand Total	**13074.24**	**11695.33**

Mode of Transport

Mode Of Transport	Count
Inter Modal	40
Pithead	138
Rail	811
Road	11
Grand Total	**1000**

Total number of power stations per state:

StateName	Number Of Power_Stations
Andhra Pradesh	6
Assam	1
Bihar	4
Chhattisgarh	15
Gujarat	4
Haryana	5
Jharkhand	5
Karnataka	4
Madhya Pradesh	8
Maharashtra	16
Odisha	4
Punjab	5
Rajasthan	4
Tamil Nadu	6
Telangana	4
Uttar Pradesh	18
West Bengal	15
Grand Total	**124**

7. Data Modelling and DAX (Power BI)

Dim_State_Table[State_Code]-> Dim_Plant_Table[State_Code]

Dim_State → Dim_Plant

Set:

- Cardinality → One to Many (1:*)
- Cross filter direction → Single
- Active → ✓

Dim_Plant->Fact_Stock

Dim_Plant_Table[Plant ID]-> Fact_Stock_Table[PlantID]

Set

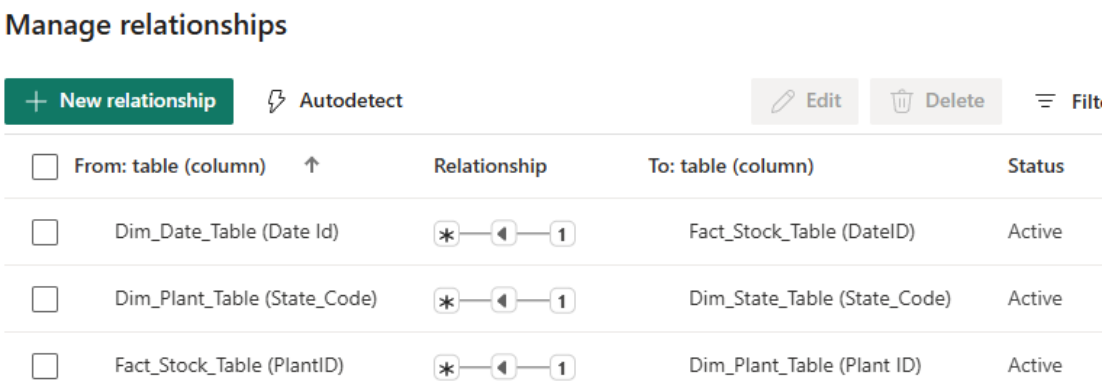
- Cardinality → One to Many (1:*)
- Cross filter direction → Single
- Active → ✓

Dim_Date → Fact_stock

Dim_Date[Date]-> Fact_Stock_Table[Date]

Set

- Cardinality → One to Many (1:*)
- Cross filter direction → Single
- Active → ✓



• Calculated Columns & DAX Measures: Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals.

Measure created to calculate Daily_Consumption_to_Daily_Req_Ratio

```
Daily_Consumption_to_Daily_Req_Ratio =
DIVIDE(sum(Fact_Stock_Table[Daily_Consumption]),sum(Fact_Stock_Table[Daily_Requirem
ent]))
```

Measure created to calculate Indegenous_to_Import_ratio

```
Indegenous_to_Import_ratio =
DIVIDE(sum(Fact_Stock_Table[Indigenous_Stock]),sum(Fact_Stock_Table[Import_Stock]))
```

Measure created to calculate the average PLF%

```
Avg PLF % = AVERAGE(Fact_Stock_Table[Plf_Prcnt])
```

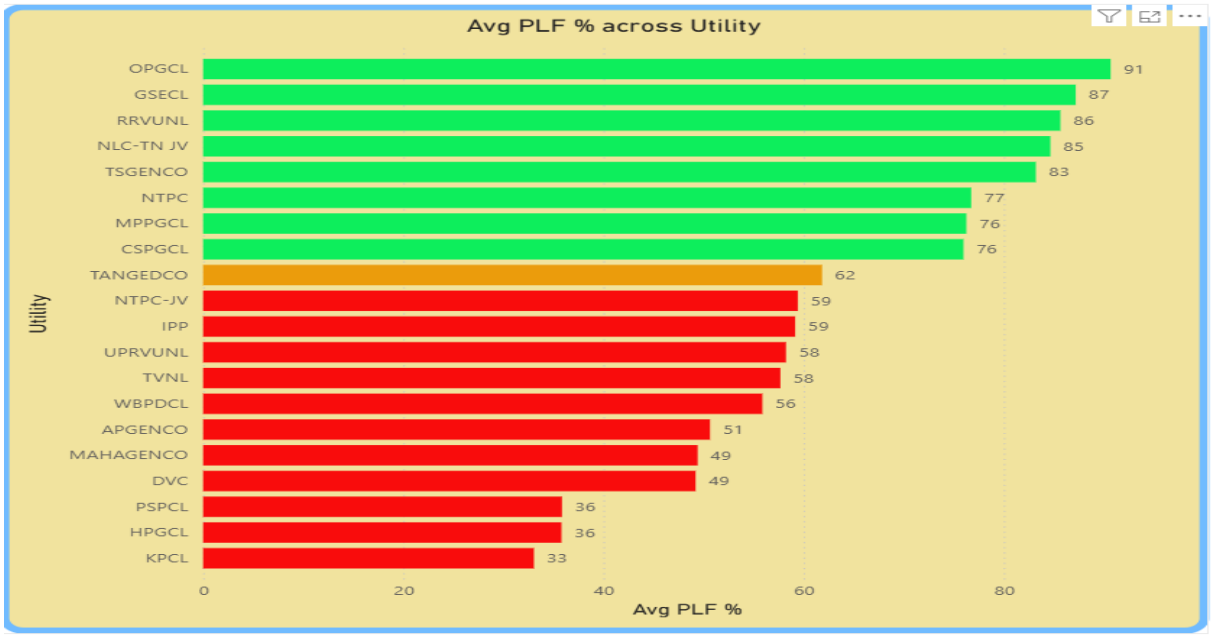
Total YTD of Import and Indigenous Stock

```
Import Stock YTD =
TOTALYTD(
    SUM(Fact_Stock_Table[Import_Stock]),
    Dim_Date_Table[Date]
)
```

```
Indigenous Stock YTD =
TOTALYTD(
    SUM(Fact_Stock_Table[Indigenous_Stock]),
    Dim_Date_Table[Date]
)
```

8. Analysis and Visualizations (Power BI)

AVERAGE PLF% ACROSS UTILITY



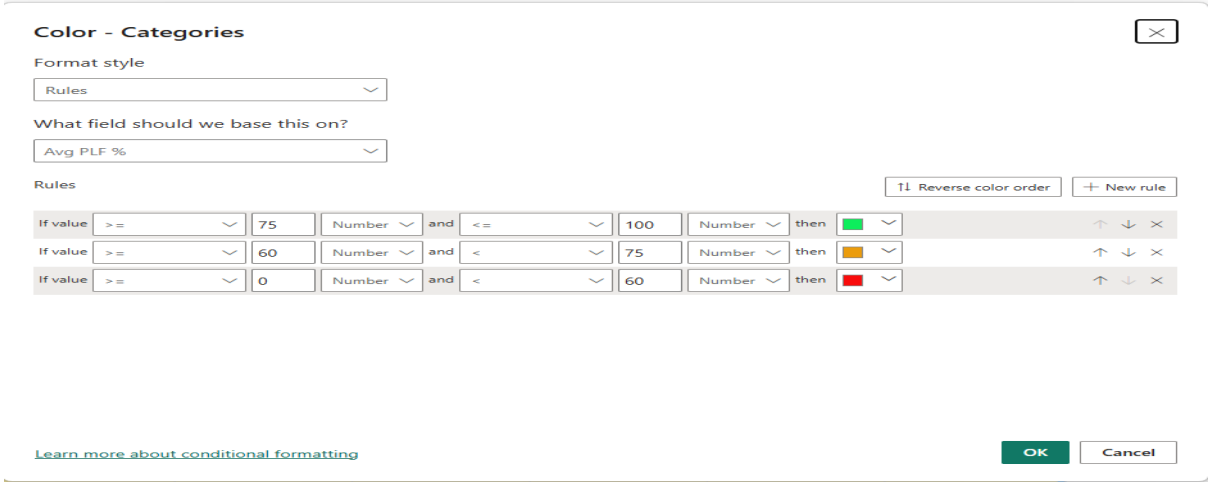
→Out of 20 utility 8 utilities are functioned effectively.

→OPGCL Utility is the effectively running utility

→KPCL,is the underperforming utility based on its PLF%

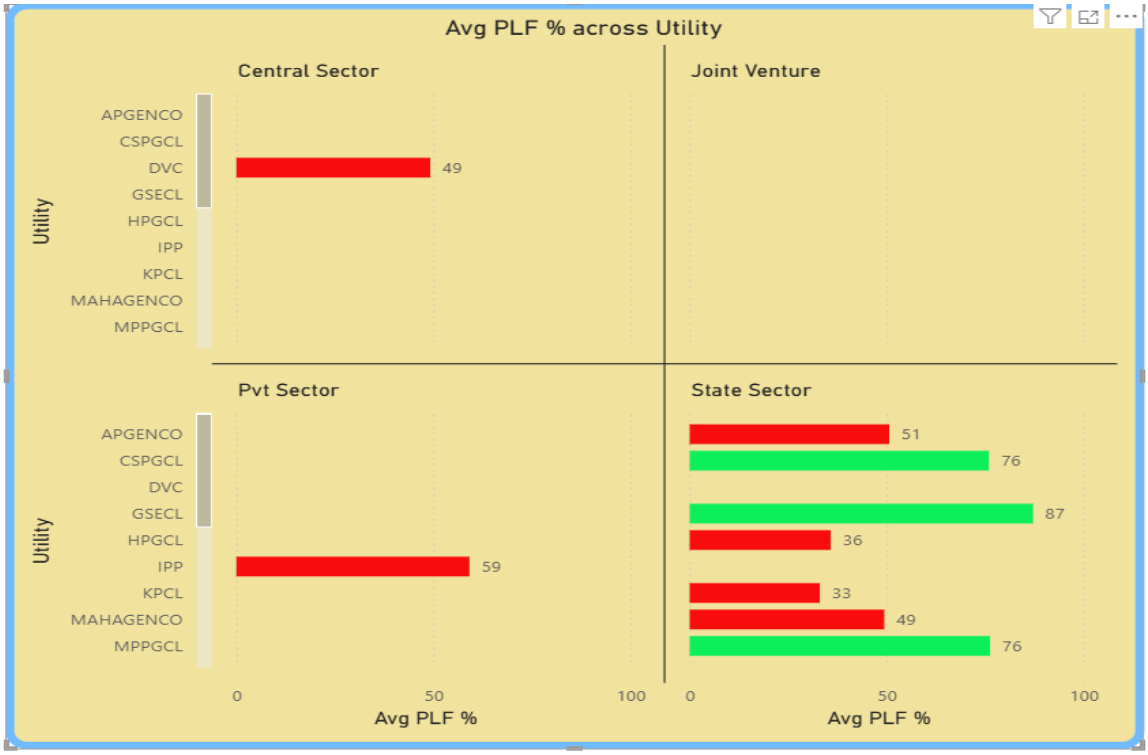
Conditional formatting rules:

In the Formating visual,click fx under Bar and then set the following conditional foramting for data colour.



AVERAGE PLF% ACROSS UTILITY SPLIT BY SECTOR

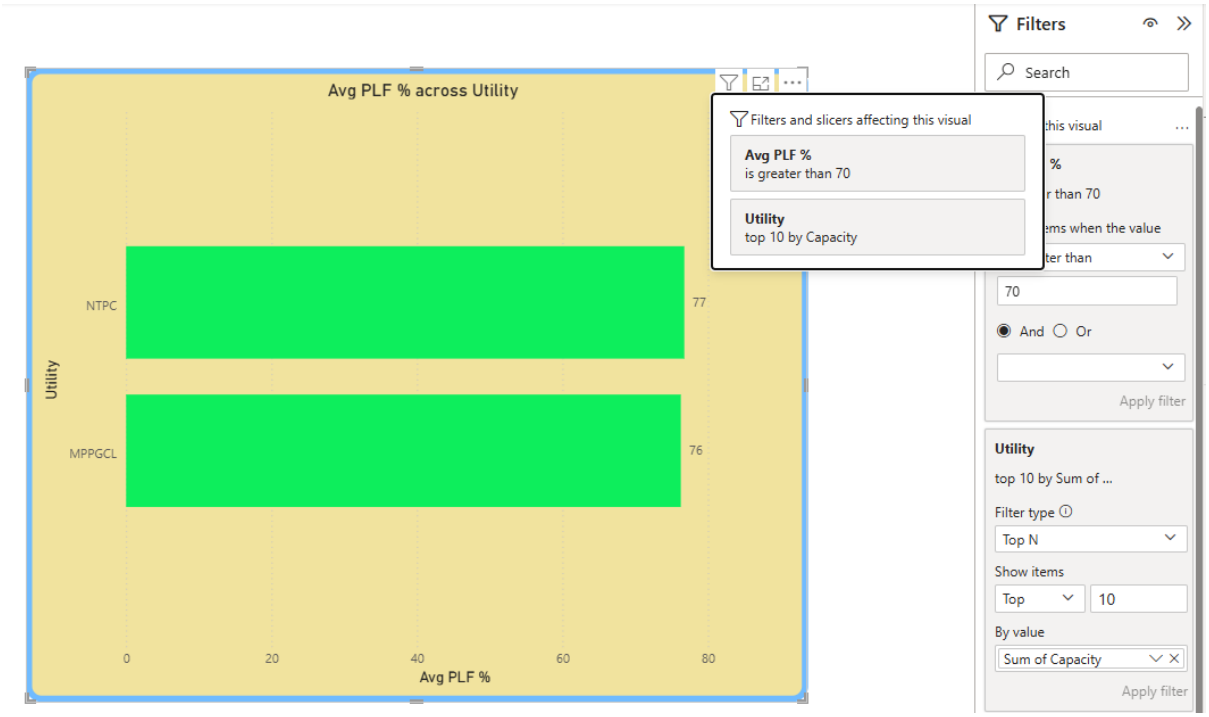
Drag Sector to small multiples we will get average PLF% by Utility split by Sector.



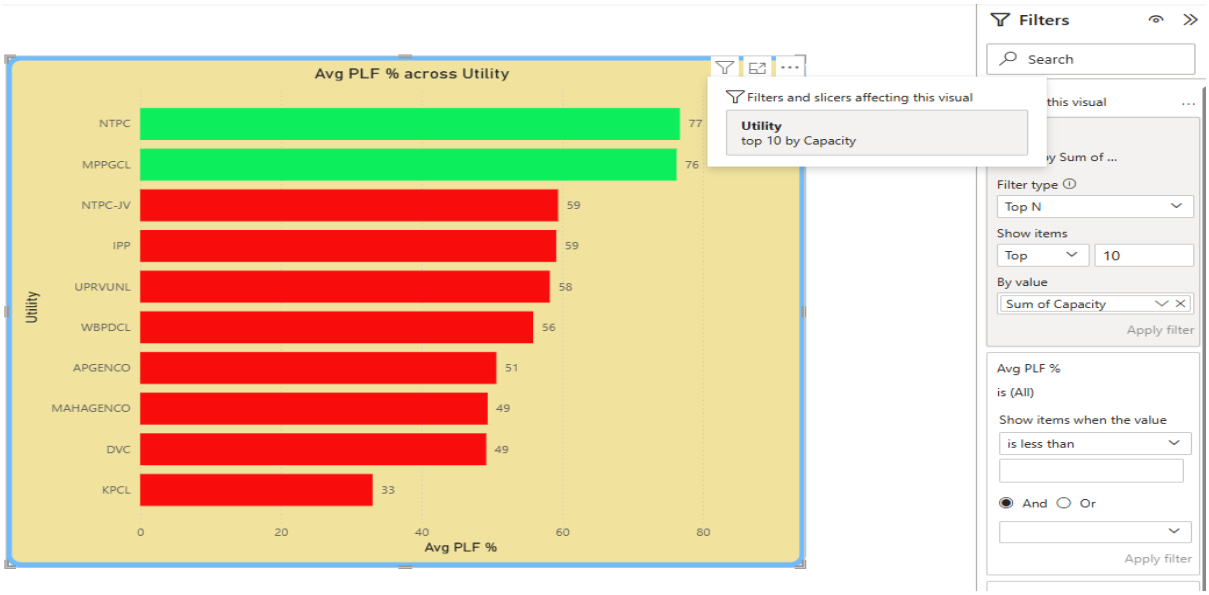
AVERAGE PLF% ACROSS UTILITY WITH FILTER

Condition 1:Avg PLF%>70 and

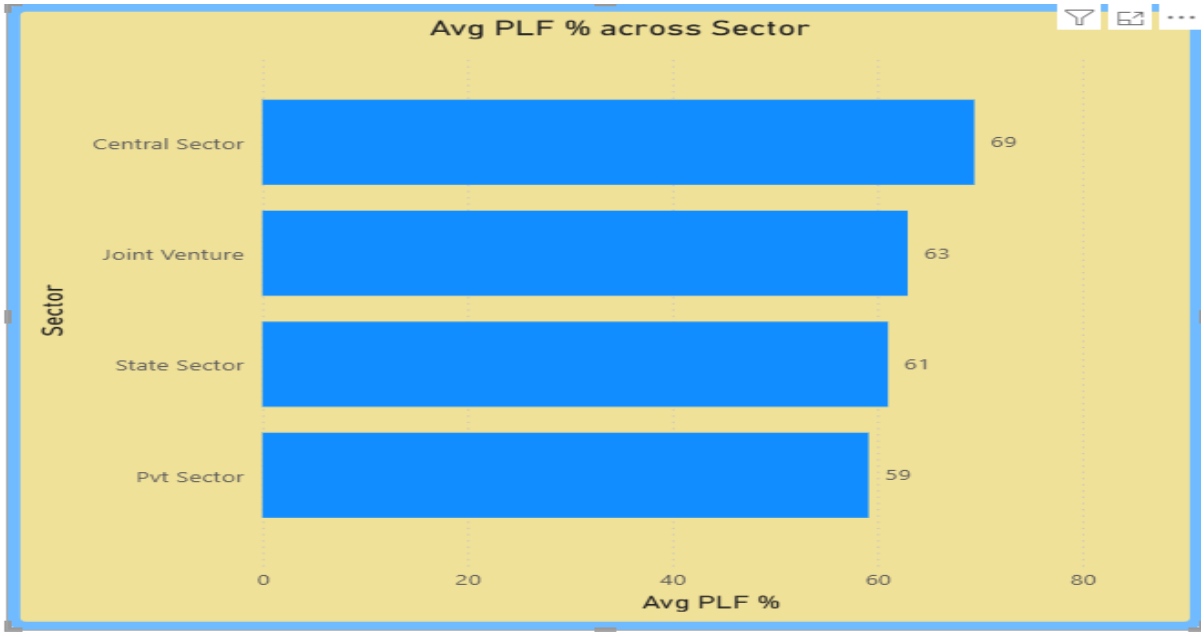
Condition 2: Top 10 by capacity



TOP 10 UTILITY BY CAPACITY



AVERAGE PLF% ACROSS SECTOR:



→Central sector has the high plf%(69) compare to all the other 3 sectors.

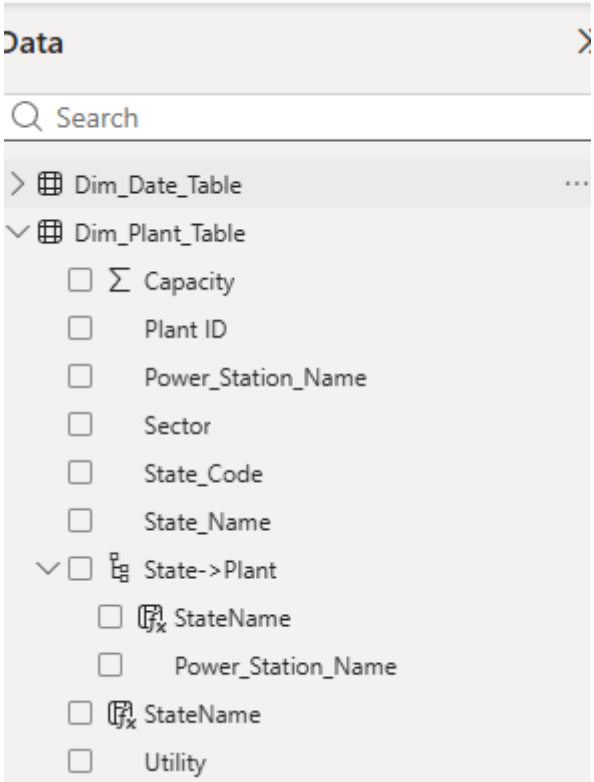
→Private sector has the lowest plf% .

Measures:

Indegenous_to_Import_ratio
22.51

Daily_Consumption_to_Daily_Req_Ratio
0.89

Drill Down



DRILL DOWN IN MATRIX:

INDIGENOUS TO IMPORT STOCK RATIO

To create a drill down hierarchy is created for state_name field in Dim_plant table .Hierarchy name is changed as State->Plant . Use this hierarchy as row value when matrix is created

Month StateName	11 Sum of Import_Stock	Sum of Indigenous_Stock	Indegenous_to_Import_ratio	12 Sum of Import_Stock	Sum of Indigenous_Stock	Indegenous_to_Import_ratio	Total Sum of Import_
Andhra Pradesh	**669.18**	**3394.56**	**5.07**	**669.18**	**3394.56**	**5.07**	**6**
Damodaram Sanjeevaiah Tps	496.45	560.14	1.13	496.45	560.14	1.13	4
Dr. N.Tata Rao Tps	0.00	1126.17		0.00	1126.17		
Painampuram Tpp	63.07	401.92	6.37	63.07	401.92	6.37	
Rayalaseema Tps	0.00	1204.48		0.00	1204.48		
Simhadri	0.00	0.00		0.00	0.00		
Vizag Tpp	109.66	101.85	0.93	109.66	101.85	0.93	1
Assam	**0.00**	**22.00**		**0.00**	**22.00**		
Bongaigaon Tpp	0.00	22.00		0.00	22.00		
Bihar	**5.58**	**627.61**	**112.47**	**5.58**	**627.61**	**112.47**	
Chhattisgarh	**14.12**	**5964.32**	**422.40**	**14.12**	**5964.32**	**422.40**	
Gujarat	**1061.89**	**2213.06**	**2.08**	**1061.89**	**2213.06**	**2.08**	**10**
Gandhi Nagar Tps	294.96	520.84	1.77	294.96	520.84	1.77	2
Sabarmati (C Station)	545.71	156.76	0.29	545.71	156.76	0.29	5
Ukai Tps	219.23	538.12	2.45	219.23	538.12	2.45	2
Wanakbori Tps	1.99	997.34	501.18	1.99	997.34	501.18	
Haryana	**0.00**	**1956.42**		**0.00**	**1956.42**		
Jharkhand	**73.86**	**6415.35**	**86.86**	**73.86**	**6415.35**	**86.86**	
Karnataka	**0.00**	**2007.28**		**0.00**	**2007.28**		
Madhya Pradesh	**239.73**	**15236.89**	**63.56**	**239.73**	**15236.89**	**63.56**	**2**
Maharashtra	**104.05**	**7545.86**	**72.52**	**104.05**	**7545.86**	**72.52**	**1**
Odisha	**135.70**	**3817.75**	**28.13**	**135.70**	**3817.75**	**28.13**	**1**
Punjab	**182.57**	**3222.84**	**17.65**	**182.57**	**3222.84**	**17.65**	**1**
Rajasthan	**82.29**	**4652.33**	**56.54**	**82.29**	**4652.33**	**56.54**	
Tamil Nadu	**1906.81**	**2940.42**	**1.54**	**1906.81**	**2940.42**	**1.54**	**19**
Total	**4477.78**	**100790.93**	**22.51**	**4477.78**	**100790.93**	**22.51**	**44**

Month	11			12		
State_Name	Sum of Daily_Consumption	Sum of Daily_Requirement	Daily_Consumption_to_Daily_Req_Ratio	Sum of Daily_Consumption	Sum of Daily_Requirement	Daily_Consumption_to_Daily_Req_Ra
Andhra Pradesh	596.95	818.27	0.73	596.95	818.27	(
Assam	21.56	41.56	0.52	21.56	41.56	(
Bihar	221.23	583.27	0.38	221.23	583.27	(
Chhattisgarh	1179.32	1605.41	0.73	1179.32	1605.41	(
Gujarat	403.38	337.43	1.20	403.38	337.43	1
Haryana	303.73	427.14	0.71	303.73	427.14	(
Jharkhand	313.77	314.30	1.00	313.77	314.30	1
Karnataka	405.36	421.44	0.96	405.36	421.44	(
Madhya Pradesh	1162.03	1236.86	0.94	1162.03	1236.86	(
Maharashtra	1726.07	1748.50	0.99	1726.07	1748.50	(
Odisha	637.21	628.35	1.01	637.21	628.35	1
Punjab	434.41	435.53	1.00	434.41	435.53	1
Rajasthan	401.61	411.73	0.98	401.61	411.73	(
Tamil Nadu	578.32	561.42	1.03	578.32	561.42	1
Telangana	564.99	564.55	1.00	564.99	564.55	1
Uttar Pradesh	1793.44	1783.80	1.01	1793.44	1783.80	1
West Bengal	951.94	1154.68	0.82	951.94	1154.68	(
Total	**11695.33**	**13074.24**	**0.89**	**11695.33**	**13074.24**	**0**

Measures and Matrixes

YTD Calculations of Indigenous and Import Stock:

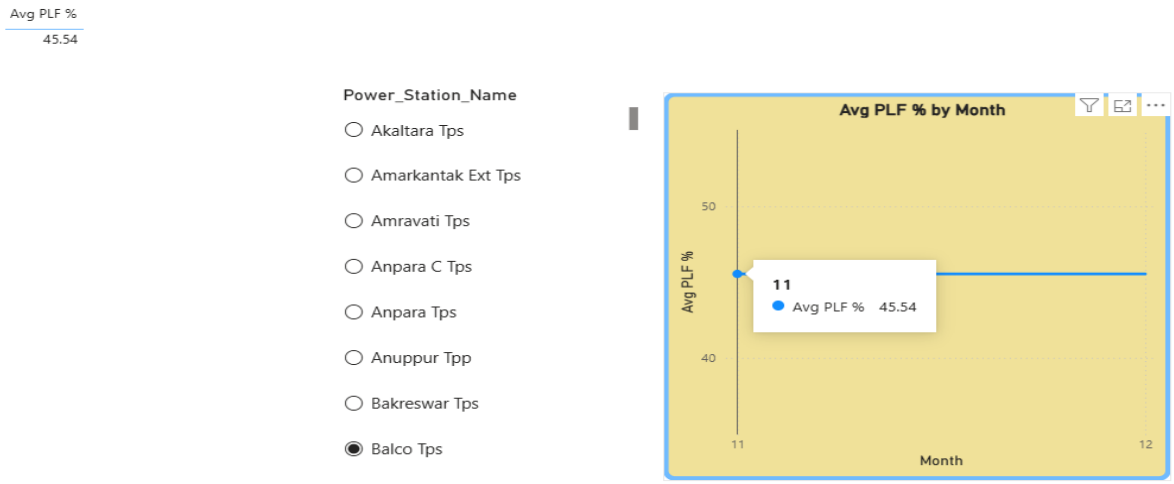
Import Stock YTD	Indigenous Stock YTD
4,477.78	1,00,790.93

State_Name	Import Stock YTD	Indigenous Stock YTD
Andhra Pradesh	669.18	3,394.56
Assam	0.00	22.00
Bihar	5.58	627.61
Chhattisgarh	14.12	5,964.32
Gujarat	1,061.89	2,213.06
Haryana	0.00	1,956.42
Jharkhand	73.86	6,415.35
Karnataka	0.00	2,007.28
Madhya Pradesh	239.73	15,236.89
Maharashtra	104.05	7,545.86
Odisha	135.70	3,817.75
Punjab	182.57	3,222.84
Rajasthan	82.29	4,652.33
Tamil Nadu	1,906.81	2,940.42
Telangana	0.00	10,662.12
Uttar Pradesh	0.00	25,627.95
West Bengal	2.00	4,484.17
Total	**4,477.78**	**1,00,790.93**

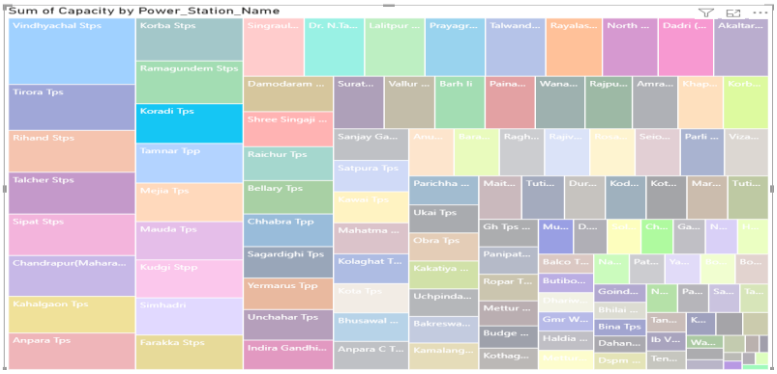
The Year-To-Date (YTD) analysis for the period of **November 27, 2018, to December 5, 2018**, highlights the cumulative consumption and stock performance across the major states:

The YTD data reveals that high consumption does not guarantee stability.
While **Uttar Pradesh** leads in volume, it faces the highest operational stress (critical events). Conversely, **Telangana** serves as the model for stability, maintaining perfect fulfillment with the highest safety reserves in the country.
Bihar remains the primary concern, showing a persistent failure to meet even 40% of its demand throughout the year-to-date period.

AVERAGE PLF% BY MONTH ACROSS POWER STATIONS WITH SLICER



STATE-WISE CAPACITY DISTRIBUTION – 124 POWER STATIONS



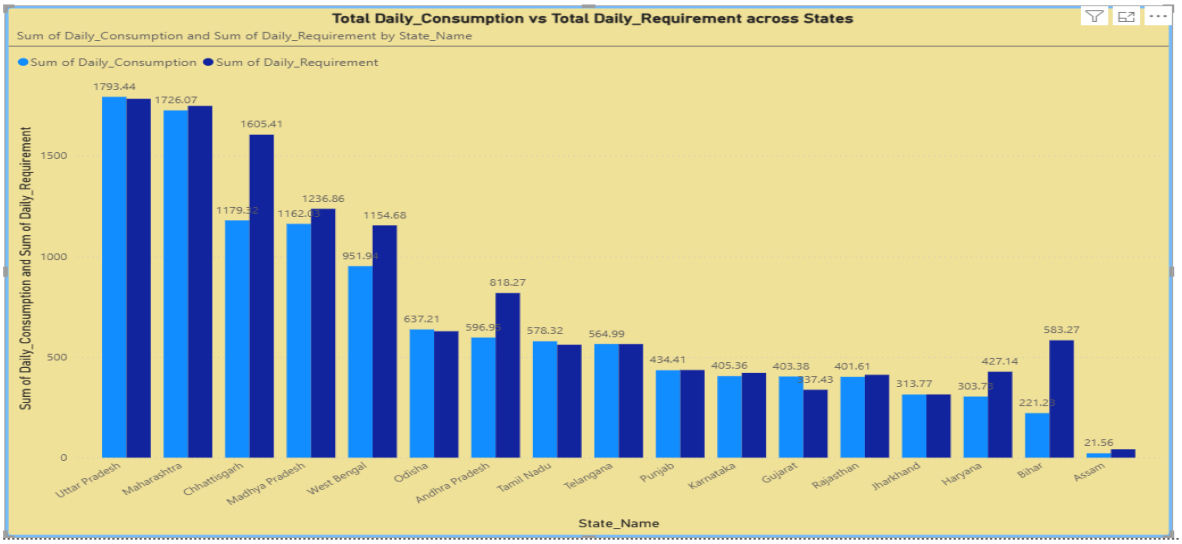
Power_Station_Name	StateName	Sum of Capacity
Koradi Tps	Maharashtra	19200
Total		19200

→Vindhyachal Stps Power Station in Madhya Pradesh has the highest capacity, 38080 MWs.

→Bandakhar Tpp Powerstation in Chhattisgarh has the lowest capacity, 600 MWs.

→Over all capacity of all the power stations is 1204398 MWs.

TOTAL DAILY CONSUMPTION VS DAILY REQUIREMENT



→ **Daily Requirement** → Coal needed per day to run at planned load (based on capacity & target PLF%).

→ **Daily Consumption** → Actual coal burned per day.

The chart shows daily coal consumption versus requirement across states.

→Uttar Pradesh leads with the highest values, while Assam has the lowest.

→ **Highest Consumption:** Uttar Pradesh (1793.44 tonnes)

→ **Lowest Consumption:** Assam (21.56 tonnes)

→States like **Punjab** and **Telangana** show bars of nearly equal height, meaning their daily needs are being precisely met by the current supply.

Here is the data for the top 5 states

→**Uttar Pradesh:** This state has a daily consumption of 1793.44 tonnes and a daily requirement of 1726.07 tonnes. It is currently in a **Surplus** status as its supply exceeds its demand.

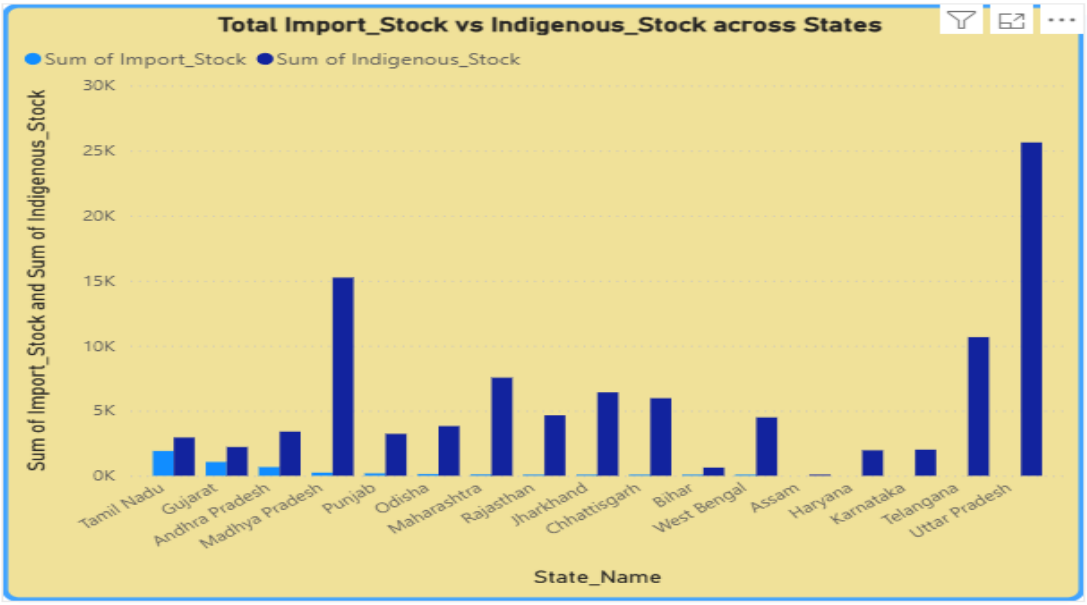
→ **Maharashtra:** The state records a daily consumption of 1726.07. Since the requirement bar is visually higher (estimated at over 1750 tonnes), it is categorized as being in a **Deficit**.

→ **Chhattisgarh:** With a daily consumption of 1179.32 against a much higher daily requirement of 1605.41 tonnes, this state is facing a **High Deficit**.

→ **Madhya Pradesh:** This state has a daily consumption of 1162.03 and a daily requirement of 1236.86 tonnes, resulting in a **Deficit** status.

→ **West Bengal:** Showing a daily consumption of 951.94 tonnes and a daily requirement of 1154.68 tonnes, this state is also in a **Deficit**.

TOTAL IMPORT STOCK VS INDIGENOUS STOCK



Indigenous stock insights:

→Across almost every state represented, **Indigenous Stock** (dark blue) significantly outweighs **Import Stock** (light blue).

→In many states, Import Stock is negligible.

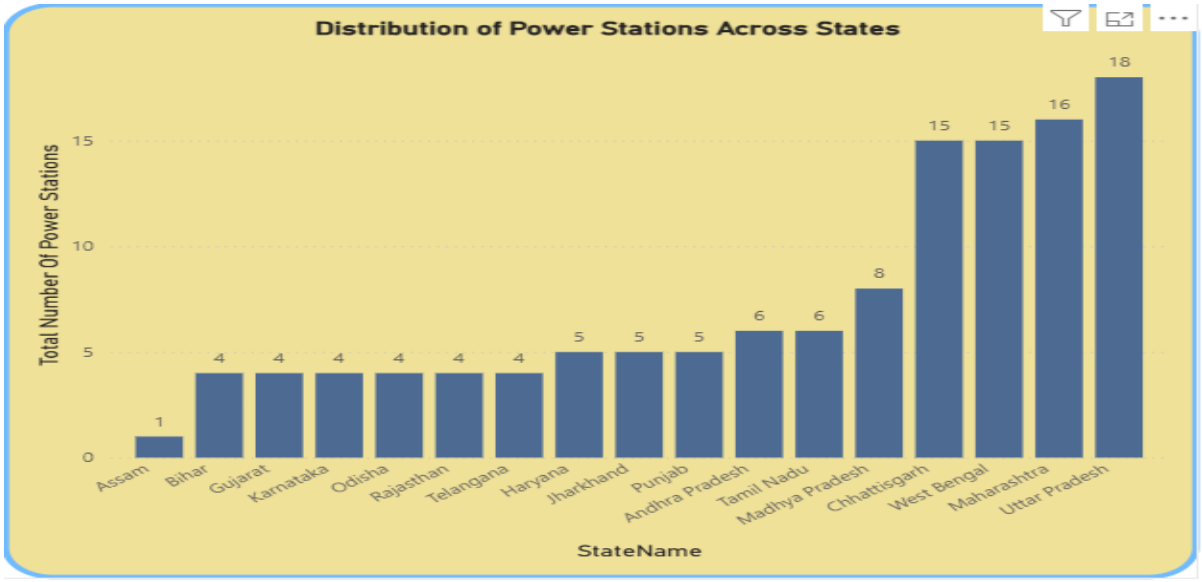
→Uttar Pradesh(25K+),MadhyaPradesh(15K+),Telengana(11k+) are the top 3 states with highest indigenous stock.

Import Stock Insights

While Import Stock is generally low, it is most visible in the following southern states:

- **Tamil Nadu:** Shows the highest volume of Import Stock (approx. **2K**).
- **Andhra Pradesh & Gujarat:** Both show small but visible amounts of imported goods compared to states like Bihar or Haryana, where it is nearly zero.

DISTRIBUTION OF POWER STATIONS ACROSS STATES



→Four states significantly lead the country in the number of power stations, forming a distinct high-volume group:

Top Tier States (15+ Stations)

- **Uttar Pradesh:** The leader with **18** power stations.
- **Maharashtra:** Follows closely with **16** stations.
- **West Bengal & Chhattisgarh:** Tied for third, each possessing **15** power stations.

Mid-Tier States (5–8 Stations)

These states show moderate distribution levels, bridging the gap between the power giants and the smaller regions:

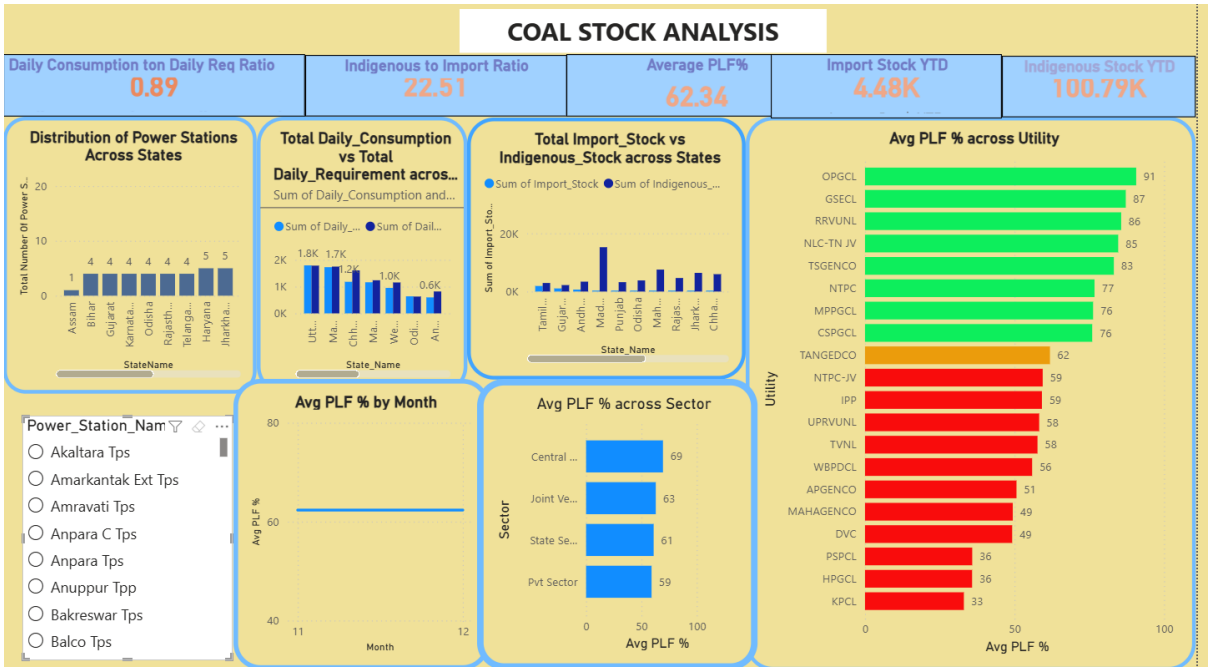
- **Madhya Pradesh:** Has **8** power stations.
- **Tamil Nadu & Andhra Pradesh:** Both have **6** stations.
- **Punjab, Jharkhand, & Haryana:** All share a count of **5** stations.

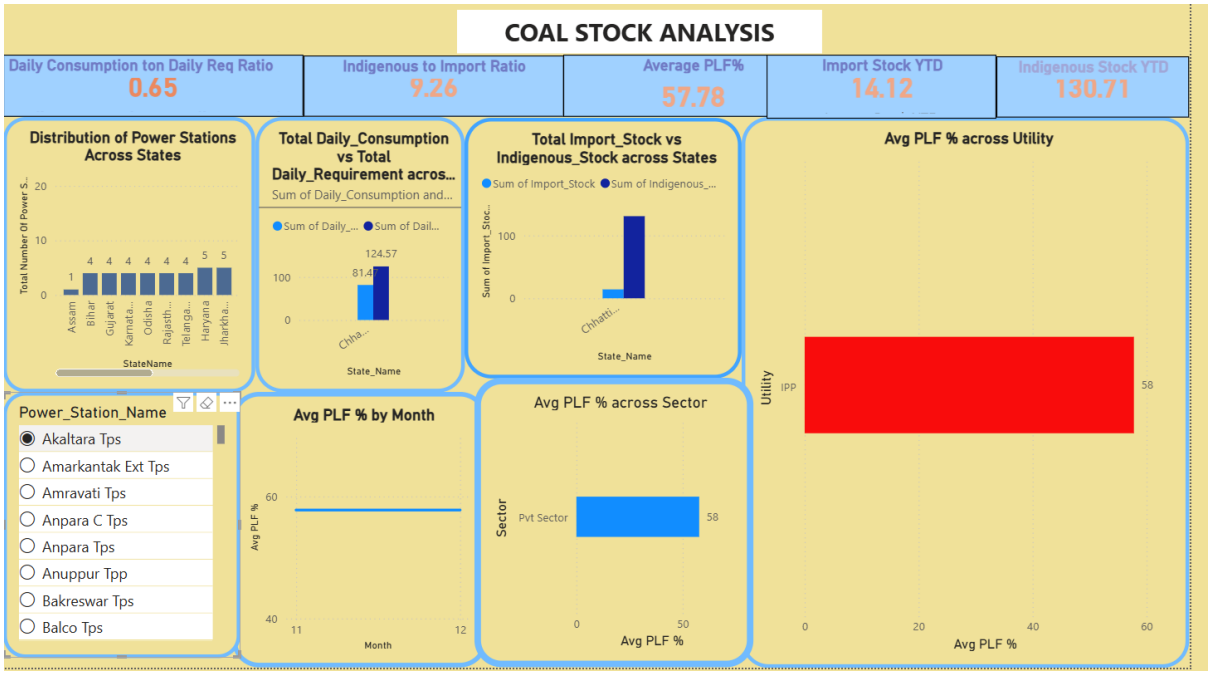
Lower Volume States (1–4 Stations)

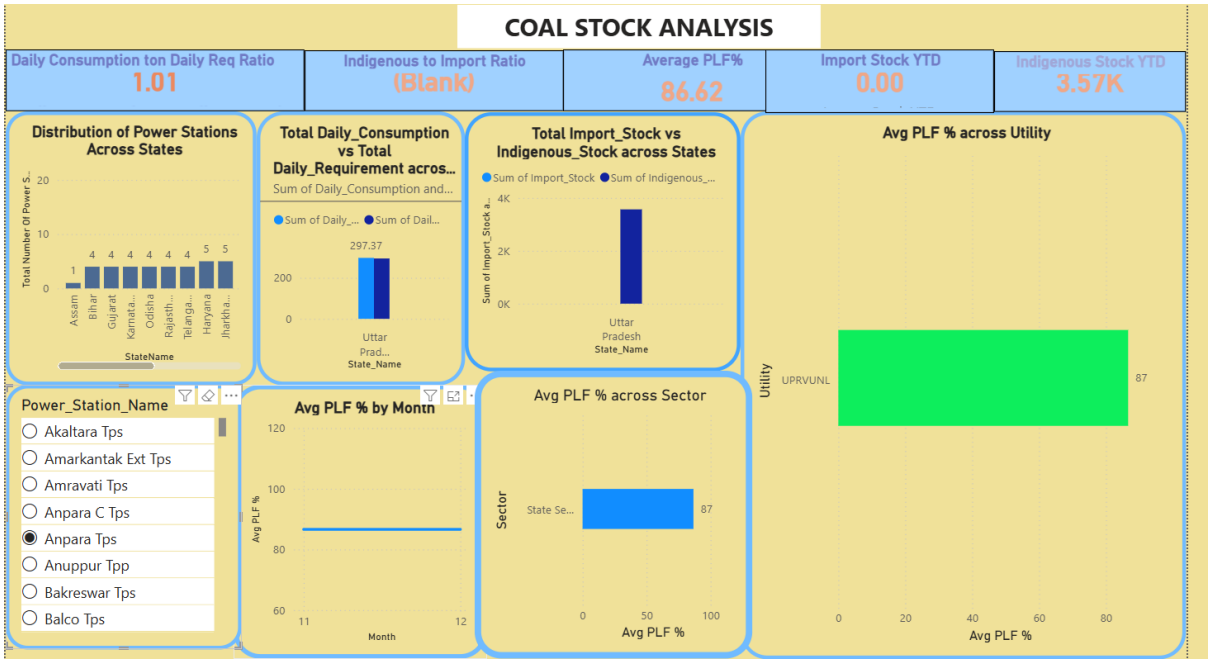
A large portion of the states listed fall into this category, indicating a more centralized power infrastructure elsewhere:

- **Cluster of 4:** Bihar, Gujarat, Karnataka, Odisha, Rajasthan, and Telangana all have exactly **4** power stations each.
- **Assam:** Represents the lowest count on the chart with only **1** power station.

SCREENSHOTS OF DASHBOARD /REPORT



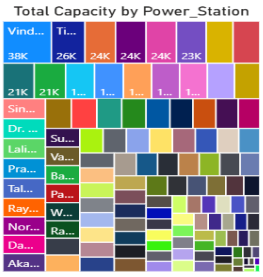




Month	11			12		
State_Name	Sum of Daily_Consumption	Sum of Daily_Requirement	Daily_Consumption_to_Daily_Req_Ratio	Sum of Daily_Consumption	Sum of Daily_Requirement	Daily_Consumption_t
Andhra Pradesh	596.95	818.27	0.73	596.95	818.27	
Assam	21.56	41.56	0.52	21.56	41.56	
Bihar	221.23	583.27	0.38	221.23	583.27	
Chhattisgarh	1179.32	1605.41	0.73	1179.32	1605.41	
Gujarat	403.38	337.43	1.20	403.38	337.43	
Total	**11695.33**	**13074.24**	**0.89**	**11695.33**	**13074.24**	

Month	11	
StateName	Sum of Import_Stock	Sum of Indigenous_Stock
West Bengal	**2.00**	**4484.17**
Uttar Pradesh	**0.00**	**25627.95**
Telangana	**0.00**	**10662.12**
Tamil Nadu	**1906.81**	**2940.42**
Rajasthan	**82.29**	**4652.33**
Punjab	**182.57**	**3222.84**
Odisha	**135.70**	**3817.75**
Maharashtra	**104.05**	**7545.86**
Madhya Pradesh	**239.73**	**15236.89**
Total	**4477.78**	**100790.93**

Total Capacity by Power_Station



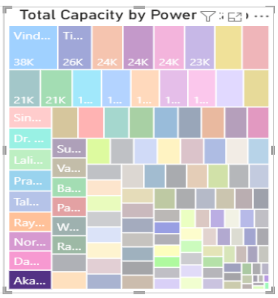
Capacity of Power stations

Power_Station_Name	StateName	Sum of Capacity
Bandakhar Tpp	Chhattisgarh	600
Barkhera Tps	Uttar Pradesh	720
Khambarkhera Tps	Uttar Pradesh	720
Kundarki Tps	Uttar Pradesh	720
Maqsoodpur Tps	Uttar Pradesh	720
Utraula Tps	Uttar Pradesh	720
Southern Repl. Tps	West Bengal	1080
Amarkantak Ext Tps	Madhya Pradesh	1680
Durgapur Tps	West Bengal	1680
Titagarh Tps	West Bengal	1920
Wardha Warora Tpp	Maharashtra	2160
Sabarmati (C Station)	Gujarat	2652
Bandel Tps	West Bengal	2760
Korba-Ii	Chhattisgarh	3040
Total		**1204398**

Month	11			12		
State_Name	Sum of Daily_Consumption	Sum of Daily_Requirement	Daily_Consumption_to_Daily_Req_Ratio	Sum of Daily_Consumption	Sum of Daily_Requirement	Daily_Consumption_t
Andhra Pradesh	596.95	818.27	0.73	596.95	818.27	
Assam	21.56	41.56	0.52	21.56	41.56	
Bihar	221.23	583.27	0.38	221.23	583.27	
Chhattisgarh	1179.32	1605.41	0.73	1179.32	1605.41	
Gujarat	403.38	337.43	1.20	403.38	337.43	
Total	**11695.33**	**13074.24**	**0.89**	**11695.33**	**13074.24**	

Month	11	
StateName	Sum of Import_Stock	Sum of Indigenous_Stoc
West Bengal	**2.00**	**4484.1**
Bakreswar Tps	0.00	243.4
Bandel Tps	2.00	233.5
Budge Budge Tps	0.00	1355.4
D.P.L. Tps	0.00	129.0
Durgapur Steel Tps	0.00	97.2
Durgapur Tps	0.00	292.6
Farakka Stps	0.00	0.0
Haldia Tpp	0.00	658.5
Total	**4477.78**	**100790.9**

Total Capacity by Power



Capacity of Power stations

Power_Station_Name	StateName	Sum of Capacity
Akaltara Tps	Chhattisgarh	14400
Total		**14400**

9. INSIGHTS

KEY FINDINGS:

- A significant national supply deficit persists, with total daily requirements of 13,074 units exceeding actual consumption of 11,695 units.
- Power demand is heavily concentrated, as the top five states account for more resource usage than the remaining twelve states combined.
- Approximately 108 power plants are flagged as "Critical," indicating their coal reserves have depleted well below mandatory safety levels.
- Bihar remains the most energy-insecure state, fulfilling less than 40% of its requirement with a critically low average stock of 1.8 days.
- West Bengal and Chhattisgarh face urgent shutdown risks, with over 100 plants holding less than 4 days of essential coal stock reserves.

1. Descriptive Analysis (What is happening?)

This analysis summarizes the current state of coal stocks and power generation metrics.

- **Total Performance:** The total daily requirement across all states is **13,074.24units**, but the actual consumption is **11,695.33 units**, indicating an overall national deficit.
- **Stock Levels:** The average coal stock across all plants is enough for **8.23 days**.
- **Criticality:** Out of 1,000 plant records analyzed, **108plants** are officially flagged as **"Critical"**, meaning their stock levels have fallen below safety norms.
- **Top Consumers:** Uttar Pradesh and Maharashtra are the largest drivers of demand and consumption in the country.

2. Diagnostic Analysis (Why is it happening?)

This analysis looks for the underlying causes of the observed gaps.

- **The Deficit Root:** The primary reason for lower consumption in states like **Chhattisgarh (-426.09 tonnes deficit)** and **Bihar (-362.04 tonnes deficit)** is a direct supply-side constraint.
- **Correlation:** There is a moderate positive correlation (**0.64 tonnes**) between Total_Stock and Daily_Consumption. This suggests that while stock availability is a major factor in how much power can be generated, other factors (like plant efficiency or transmission constraints) also play a role.
- **Under-performance in Bihar:** Bihar shows a massive deficit because its consumption is less than half of its requirement, likely due to having the lowest average stock days (**1.8 days**) in the entire dataset.

3. Predictive Analysis (What is likely to happen?)

This analysis identifies future risks and trends based on current data patterns.

- **High-Risk Zones: West Bengal** and **Chhattisgarh** are at high risk of potential power outages or plant shutdowns. West Bengal has **61 plants** with less than 4 days of stock remaining, and Chhattisgarh has **48**.
- **Supply Depletion:** If current consumption patterns continue without immediate replenishment, the 108 critical plants are likely to reach zero stock within **1 to 3days**, leading to a significant drop in national power grid stability.
- **Widening Gap:** In states like Haryana and Bihar, the gap between requirement and consumption is likely to widen as their stock levels are insufficient to support even current lower-than-required consumption.

4. Prescriptive Analysis (What should we do?)

This analysis provides actionable recommendations to solve the identified problems.

- **Immediate Redirection:** Priority for coal shipments should be shifted immediately to **Bihar**, which has both a high deficit and a critically low stock average of only **1.8 days**.
- **Logistics Optimization for Chhattisgarh:** As the state with the highest absolute deficit (426.09), a dedicated supply corridor or increased "Rail" transport (the primary mode) should be allocated to stabilize its industrial demand.
- **Stockpile Minimums:** Policy should be updated to mandate a minimum of **7 days of stock** for plants in high-demand states like West Bengal and Maharashtra to buffer against supply chain disruptions.
- **Efficiency Audit:** For plants with high stock but low consumption, an operational efficiency audit is recommended to determine if technical failures are preventing the usage of available coal.

10.CONCLUSION

- **Systemic Supply Gap:** India faces a significant energy shortfall as the total daily consumption (11,695.33 tonnes) fails to meet the national requirement (13,074.24).
- **Operational Vulnerability:** With 108 plants in "Critical" status and over 100 plants in high-risk zones holding less than four days of fuel, the power grid operates on a razor-thin safety margin.
- **Severe Regional Disparity:** A drastic divide exists between surplus-performing states like Uttar Pradesh and crisis-bound states like Bihar, which fulfills less than 38%of its daily demand.
- **Imminent Shutdown Risks:** The concentration of low-stock plants in West Bengal and Chhattisgarh indicates a high probability of localized blackouts without immediate logistical intervention.
- **Strategic Requirement:** To stabilize the grid, the analysis mandates an immediate prescriptive shift in coal logistics to prioritize high-deficit states and replenish critical reserves.